

Pavani Kuppili

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Research focus: Programmable data planes; datacenter networking; FPGA SmartNICs and DPUs; P4; RDMA, DPDK, XDP/eBPF; network measurement and hardware offload.

Education

University of Utah — Ph.D. in Computer Science Aug. 2023–Present
Flux Research Group; Advisor: Prof. Robert Ricci; GPA: 3.89/4.00 Salt Lake City, UT
National Institute of Technology Silchar — B.Tech., Electronics & Communication Eng. 2015–2019

Research Experience

University of Utah — Flux Research Group Aug. 2023–Present
Graduate Research Assistant Salt Lake City, UT

- Developed **p4cketnet**, a P4/Tofino datacenter-network emulator with 406-ns per-hop delay granularity at near line rate; used it to study RDMA collectives in distributed ML training.
- Built and evaluated matched *SmartRetry* XDP and P4 FPGA-NIC retry-admission prototypes at 300 kpps; both admitted approximately 72 krequests/s while the FPGA moved filtering off the host CPU.
- Characterized RDMA on-demand paging: nonresident small reads incurred an approximately 2.1-ms floor, while concurrent faults to one page coalesced into one storage read; the resulting poster won Best Poster at N2Women, SIGCOMM 2026.

University of Massachusetts Amherst Summers 2024, 2025, 2026
Cloud Testbed Development Intern; Advisors: Profs. Michael Zink and Miriam Leeser Amherst, MA

- Built *LatencyLab*, a DPDK framework for measuring P4-pipeline latency on AMD Alveo U280 FPGA SmartNICs; across four programs, p99 stayed within 22 ns of the median and medians reproduced within 1–2 ns across five bring-ups (IEEE HPEC 2026).
- Developed P4 packet-processing pipelines and OpenNIC Shell testbed support for reproducible, line-rate FPGA SmartNIC experiments in the Open Cloud Testbed.

Selected Publications and Posters

- **P. Kuppili**, Y. Qian, S. Handagala, M. Zink, M. Leeser, and R. Ricci. “LatencyLab: A DPDK-Based P4 Pipeline Latency Measurement Framework for FPGA SmartNICs.” *IEEE HPEC*, 2026 (accepted). [Code and data](#).
- **P. Kuppili**, A. Maricq, B. E. Stephens, R. Stutsman, and R. Ricci. “Poster: TopoCloud: Getting Datacenter Network Experiments Into the Right Shape.” *IEEE ICNP*, 2024.
- “Can RDMA Applications Use Page-Fault Information Before a Read?” Research poster, N2Women at ACM SIGCOMM 2026. **Best Poster Award**.

Industry Experience

Amazon, India May 2019–July 2022
Systems Development Engineer, Global Websites Cloud Hosting Jan. 2022–July 2022

- Automated scaling operations in JavaScript, resolving 70%+ of related on-call tasks; executed the global live migration of Amazon retail websites from legacy load balancers to AWS ELB.

Cloud Engineer, AWS EC2 Linux May 2019–Jan. 2022

- Diagnosed a critical AWS SSM Agent defect and enabled custom-OS log integration with CloudWatch; resolved 1,000+ customer cases spanning Linux, EC2 performance, and multi-tier AWS architectures.

Technical Skills

Data planes and systems: P4, Intel Tofino, AMD Vitis Networking P4, Alveo U280, OpenNIC Shell, DPDK, RDMA, XDP/eBPF, Linux, AWS, Docker, CloudLab/OCT, Wireshark.

Programming: C, Python, Bash/Shell, JavaScript, Go, SQL, Java, CUDA, MPI, OpenMP, L^AT_EX.

Honors and Service

Awards: N2Women Best Poster and Travel Grant, SIGCOMM 2026; Travel Grants, ICNP 2024 and EuroSys 2024; AWS Quarter Superstar, 2020; AWS Certified Solutions Architect–Associate, 2020.

Service: IMC 2026 Shadow TPC; Artifact Evaluation Committees for OSDI, CoNEXT, and ICPE 2026; EuroSys 2024 Shadow PC.